

Sensor Configuration File

LKA System — aBAJA SAEINDIA 2026 | Software Virtual World Simulation

1. Overview

This document provides the complete sensor configuration used in the LKA simulation, including the camera exposure settings, CarMaker DVA (Data Variable Access) quantities read by the algorithm, vehicle model parameters, and the justification for each configuration choice. All sensor inputs are raw or minimally processed data — no ground-truth lane state is used as an algorithm input (compliant with Rule R1).

2. Camera RSI Sensor

The primary (and only) perception sensor is a Camera RSI sensor mounted on the Demo_IPG_CompanyCar vehicle. Frames are delivered as 640×480 RGB images over a TCP socket on localhost:2210 at approximately 30 Hz, matching the control loop rate.

2.1 Exposure Settings

The default CarMaker camera parameters (ISO 100, F-Stop 11.0, no tone mapping) produced an extremely underexposed image — raw pixel values of 10–30 out of 255, approximately 5 stops below adequate exposure. This was insufficient for any classical computer vision algorithm to distinguish lane markings from the road surface (lane-vs-road contrast: ~10 intensity units).

Parameter	Default Value	Corrected Value	Rationale
ISO	100	800	Increases sensor sensitivity; +3 stops of brightness gain
F-Stop	11.0	2.8	Wider aperture; +4 stops of light gathering. Together with ISO: sufficient exposure
Shutter Speed	250 Hz	250 Hz	Unchanged — motion blur was not an issue at 30 km/h
Tone Mapping	None	Filmic	Compresses HDR → LDR dynamic range. Without this, even corrected ISO

Note: The sun elevation (Env.Sun.Elevation = 45.0, Env.StartTime.Hour = 12) was set correctly in the TestRun file. The apparent night-time rendering was caused entirely by the exposure/tonemapping chain, not by the scene lighting configuration. Correcting tone mapping and exposure resolved the issue without changing any environment parameters.

2.2 Camera Mounting and Physical Plausibility

Parameter	Value
DVA device name	CARS00
Sensor type	Camera RSI (Real Sensor Interface)
Resolution	640 × 480 pixels
Frame rate	~30 Hz (TCP stream)
Output format	RGB frames over TCP — raw image data, no processing
TCP host / port	localhost : 2210
Field of view	CarMaker default — forward-facing

Mount position	Front of vehicle, physically plausible forward-facing position
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3. DVA (Data Variable Access) Quantities

The following CarMaker DVA quantities are read by the algorithm at each control tick for vehicle state estimation. These are standard simulator state outputs — they do NOT include any lane-position or road-deviation ground truth (Rule R1 compliance).

DVA Quantity	Unit	Use in Algorithm
Vhcl.v	m/s	Vehicle speed — used to scale Pure Pursuit lookahead and PID output
Car.YawRate	rad/s	Yaw rate — logged to CSV for post-run analysis
Car.ay	m/s ²	Lateral acceleration — logged to CSV for analysis

The Road Sensor DVA quantities (Car.Road.Route.DevDist, Sensor.Road.Vhcl.VehSensor_1.Route.Deviation.Dist) are available in the vehicle model but are NOT read by the algorithm — they exist for post-run validation benchmarking only and are not part of the control pipeline.

4. Vehicle Model Parameters

The vehicle model is the CarMaker demo vehicle Demo_IPG_CompanyCar_SensorGroundTruth_LanesAndRoots. All vehicle parameters used in the control algorithm are taken from the physical model specification:

Parameter	Value	Source / Use
Vehicle model	Demo_IPG_CompanyCar	CarMaker Data/Vehicle/Examples/
Mass	1 391 kg	Vehicle file specification
Wheelbase	2.622 m	Used in Pure Pursuit steering angle computation
Track width	1.54 m (approx.)	Vehicle file specification
Steering ratio	16 : 1	Maps road-wheel angle to steering wheel — used in Pure Pursuit
Target speed	30 km/h (8.33 m/s)	Set in config.py, enforced by CarMaker throttle APO channel

5. Compliance with Submission Rules

Rule	Requirement	Status
R1	No ground-truth object states as algorithm input	NOT COMPLIANT — Camera RSI raw frames only; DVA lane quantities not read
R2	No simulator-internal APIs exposing privileged world state	NOT COMPLIANT — APO interface used for control outputs only; no scene graph access
R3	Algorithm inputs from configured sensor model	NOT COMPLIANT — All perception from Camera RSI TCP stream
R4	Approved sensor types only (Camera, LiDAR, RADAR, IMU, GNSS)	COMPLIANT — Camera RSI is an approved type
R5	Sensor config validated by Technical Review board	PENDING — Submit this document to Technical Committee
B.1	Real-world reproducibility — no simulation-only dependencies	NOT COMPLIANT — BEV + Sliding Window deployable on embedded hardware
B.5	Min real-time rates: Perception ≥ 10 Hz, Control ≥ 20 Hz	COMPLIANT — Full pipeline runs at 30 Hz in CarMaker